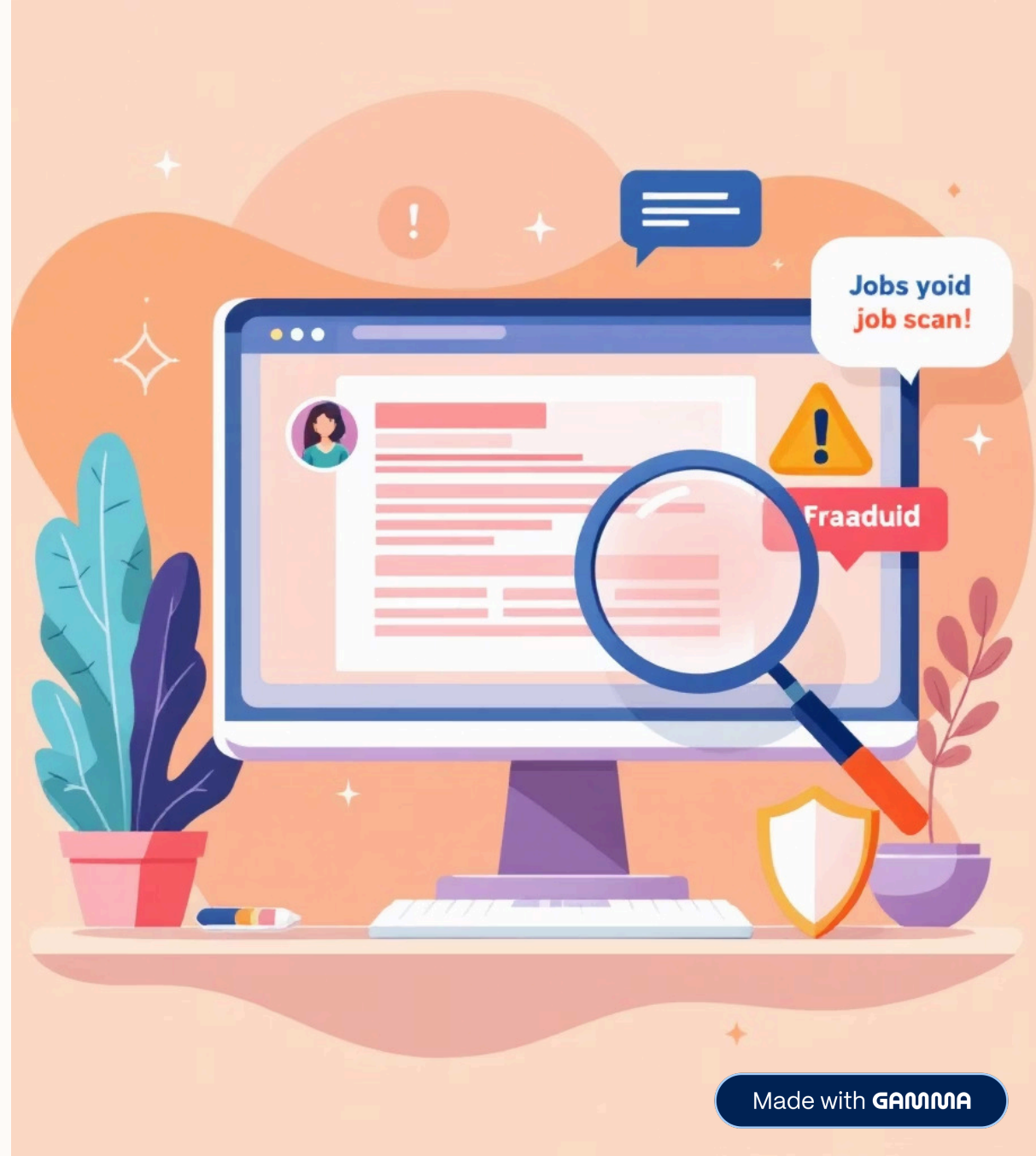


Fraudulent Job Posting Detection using Machine Learning

Author: Martín Paz



Problem & Objective

Problem

Fake job postings can lead to scams, identity theft, and loss of trust in job platforms.

Objective

Build a supervised ML model to classify job offers as:

- ✓ Real (0)
- ✓ Fraud (1)

Success Focus (Fraud Detection)

In fraud detection, the priority is:

- ✓ **High Recall** (catch as many frauds as possible)

Dataset Overview

Dataset & Target

Job postings dataset loaded from GitHub, containing real and fraudulent job advertisements.

Target column: fraudulent (binary label)

- 0 = Legitimate job posting
- 1 = Fraudulent job posting

Text Features (Unstructured Data)

These columns contain natural language and require NLP preprocessing:

- **title** → job role name
- **description** → main job content (tasks, responsibilities)
- **requirements** → skills, qualifications, expectations

Binary Features (Structured Data)

Already numeric (0/1) and help add context:

- **telecommuting** (0/1) → remote job indicator
- **has_company_logo** (0/1) → if the post includes a company logo
- **has_questions** (0/1) → if the post includes screening questions

	title	location	description	requirements	telecommuting	has_company_logo	has_questions	fraudulent
0	Architect (Middleware - MQ) - Kuwait	KW, KU,	On behalf of our client, a well known multinat...	-Working technical knowledge of IT systems and...	0	1	0	0
1	Interviewing Now for Sales Rep Positions -- wi...	US, TX, Corpus Christi	We are Argenta Field Solutions, a rapidly expa...	#NAME?	0	1	0	0
2	Process Controls Staff Engineer - Foxboro I/A ...	US, TX, USA Southwest	Experienced Process Controls Staff Engineer is...	At least 10 years of degreed professional expe...	0	0	0	0
3	Experienced Telemarketer Wanted - Digital Solu...	AU, NSW,	If you have a passion for people and love to s...	Responsibilities - Prospecting, following up a...	0	1	0	0
4	Senior Network Engineer	GB, ENG, London	As the successful Senior Network Engineer you ...	Essential skills:•Juniper switching/routing/se...	0	1	0	0

Data Cleaning & Text Preprocessing

Text Cleaning

- lowercase conversion
- remove special characters (letters only)
- normalize extra spaces

Feature Engineering

Created unified text feature:

text = title + description + requirements

Convertin text TF-IDF

Convert text into numerical features.

TF-IDF assigns a score to each word based on:

This helps the model focus on important and meaningful words instead of very common ones.

Extra numeric features

Besides text, we also included numeric features:

- telecommuting
- has_company_logo
- has_questions

We applied scaling to these numeric features so they integrate properly with the TF-IDF output.

Models Tested

01

Logistic Regression

Tested with different thresholds

02

Decision Tree

03

Random Forest (Ensemble Model)

Basic

Tuned (GridSearchCV)

Logistic Regression

Model Overview

The model uses Logistic Regression to build a strong baseline classifier that works **well with high-dimensional sparse features**. It applies the same preprocessing pipeline (TF-IDF for text + binary features).

Variant 1 - Threshold 0.5 (Default)

- Recall: **0.80**
- Precision: **0.47**
- Accuracy: **0.94**
- Frauds missed: **18** out of 91

Variant 2 - Threshold 0.3 (Optimized)

- Recall: **0.91**
- Precision: **0.30**
- Accuracy: **0.89**
- Frauds missed: **8** out of 91

Key Insight

Lowering the threshold helps catch more fraud cases, but it also creates more false alarms. This is usually **acceptable in fraud prevention** because it is better to flag suspicious posts than to miss real fraud.

Decision Tree Classifier

Model Overview

- The model uses a **Decision Tree** to capture non-linear patterns in the data. It applies the same preprocessing pipeline (**TF-IDF for text + binary features**)

Key Insights

- Inferior to Logistic Regression for text-based fraud detection.
- Cannot see the "big picture" of fraudulent language
- It looks at one word at a time to make decisions with 20,000 different words, it gets confused

Performance

- Recall: **0.77**
- Precision: **0.25**
- Accuracy: **0.87**
- Frauds missed: **21** out of 91

Random Forest (Ensemble Model)

Model Overview

- Random Forest classifier to capture complex and non-linear relationships in the data by combining multiple decision trees, in this case, ensemble of 300 decision trees to reduce overfitting and improve robustness
- Tuned. It is the **same Random Forest model**, but improved by testing different settings automatically using **GridSearchCV**. The tuned Random Forest tests more parameter settings and automatically selects the best combination.

Key Insights

- Too conservative, misses over 50% of frauds even after tuning. Sparse TF-IDF data prevents effective ensemble learning.

Baseline Configuration

- Recall: **0.47**
- Precision: **1.00**
- Accuracy: **0.97**
- Frauds missed: **48** out of 91

Tuned with GridSearchCV

- Recall: **0.54**
- Precision: **0.89**
- Accuracy: **0.97**
- Frauds missed: **42** out of 91

Model Comparison + Winner

Final Model Comparison (Fraud-focused)



Logistic (0.5): Recall 0.80 | Missed fraud = 18



Logistic (0.3): Recall 0.91 | Missed fraud = 8 ✓



Decision Tree: Recall 0.77 | Missed fraud = 21



Random Forest baseline: Recall 0.47 | Missed fraud = 48



Random Forest tuned: Recall 0.54 | Missed fraud = 42



Logistic Regression with threshold 0.3

Reason: Highest recall + lowest missed fraud cases.

Conclusion & Future Work



Key Takeaways

- ✓ Logistic Regression performs best, however not super accurate.
- ✓ Threshold tuning was more effective, so depending on case it is more important to have more false calls but better fraud detection.
- ✓ In fraud detection: **Recall > Precision**



Future Work

- Test advanced models, assure and improve the accuracy.
- Probability calibration
- More trees
- Texts treatment

Thanks!